Outlook

[IEICE] Notification of Review Evaluation 2025EDP7269

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[Notification of Rejection]

June 03, 2026.

Manuscript ID: 2025EDP7269
Paper/Letter: Paper
Issue: ED
Regular/Special: Regular
Title of Special Section:
Title: Compactness of Finite Unions of Regular Patterns and Regular Patterns without Adjacent Variables
Date of Evaluation: June 02, 2026.
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!! ATTENTION !!

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https://review.ieice.org/judgement/judgement_receipt_e.aspx?lid=2025EDP7269&code=2606031800581

Dr. Tomoyuki UCHIDA
Hiroshima City University

Dear Dr. Tomoyuki UCHIDA,
We regret to inform you that your above-referenced manuscript

has been rejected for publication in the IEICE Transactions.

Having carefully reviewed and evaluated your manuscript, we have come to the conclusion that your approach is not sufficient to warrant its publication this time.

Thank you for considering the IEICE Transactions to present the results of your research. We hope that this decision will not discourage you from submitting other manuscripts to us in the future.

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Sincerely yours,

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Associate Editor

First of all, I sincerely apologize for the significant delay in reporting the review results.
AE has received the review results from two reviewers.
Based on the review results, AE has decided that this paper is rejected.

As Reviewer A pointed out, many of the papers cited in this paper are older, and their novelty from the perspective of the latest research trends is unclear.
Furthermore, since this paper is very long, it would be desirable to shorten it by selecting the most relevant content to improve readability.

Reviewer A

The paper studies substitution function on sequences consisting of constants and variables.
The paper is so long that it is hard to read through it, and most readers might not obtain enough useful information. The paper might be interesting for theoreticians. However, the effectiveness of the result is doubtful as the contribution on computational learning theory is only based on old references.

Proof of the lemma 4 seems to have a flaw. It assumes the case p<q does not hold (p.6 left l.10) but p<q follows from the argument (p.6 left l.24).

I cannot get the significance of figures 9, 11, 13 and so on.

Reviewer B

This paper studies the compactness property with respect to language containment for finite unions of regular pattern languages. The authors revisit the results of Sato et al.[4], point out an error in a lemma used in their proof by presenting counterexamples, and provide a revised proof under additional conditions. Furthermore, the paper introduces a subclass of regular patterns, called non-adjacent regular patterns, and establishes characteristic-set and compactness results for this subclass under weaker assumptions on the alphabet size.

The topic is relevant to pattern language theory and computational learning theory. In particular, the correction of a previously published result and the investigation of a restricted class of regular patterns are potentially valuable contributions.
However, I consider that several aspects of the manuscript require

clarification and improvement, as follows.

1.
The paper presents counterexamples to Lemma 14 of Sato et al.[4]. While the counterexamples appear to demonstrate that the lemma does not hold in general, the manuscript focuses primarily on showing the failure of the lemma itself. It would be beneficial to explain more explicitly which step of the original proof becomes invalid and what structural property of the counterexamples causes the failure. Such a discussion would help readers understand the necessity of the newly introduced conditions.

2.
The manuscript introduces additional conditions in order to repair the incorrect lemma. However, it is not entirely clear whether the resulting theorem has the same scope as the original result or whether the additional assumptions effectively restrict the original problem setting. The relationship between the original lemma and the corrected version should be discussed more explicitly.

3.
The introduction of the subclass of non-adjacent regular patterns constitutes one of the main contributions of the paper. However, the motivation for focusing on this subclass is not sufficiently explained. The authors are encouraged to discuss why this restriction is natural or important from the viewpoint of pattern learning and to clarify its relationship to previously studied restricted classes of pattern languages.

4.
The manuscript contains a large number of lemmas, propositions, and lengthy case analyses. Although each result is proved in detail, the relationships among these intermediate results are not always clear. A roadmap explaining how the lemmas contribute to the subsequent propositions and main theorems would substantially improve

readability. In particular, the logical structure of Section 3 is difficult to follow because many case analyses are presented consecutively.

5.
Several concepts and notations are introduced in the Introduction and then defined again in the Preliminaries section. While some overlap is unavoidable, the presentation could be streamlined by reducing detailed formal definitions in the Introduction and leaving them to the Preliminaries section.
Furthermore, Table 1 and Table 2 appear to be nearly identical. Please check whether this duplication is intentional and revise the presentation if necessary.

6.
The manuscript combines two substantial contributions: (i) correction of a previous compactness result and (ii) compactness properties of non-adjacent regular patterns. While these topics are related, the overall focus of the paper is somewhat diffuse. The authors may consider clarifying the connection between these contributions and emphasizing more clearly the primary message of the paper.